

Lead Scoring Model for X Education

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Problem Statement

- X Education faces a low lead conversion rate (~30%) despite high lead volume.
- Objective: Identify 'Hot Leads' with high conversion potential.
- Goal: Build a model that predicts conversion likelihood and improves efficiency of the sales team.

Dataset Overview

- Source: Historical lead data (~9000 records)
- Target Variable: Converted (1 = converted, 0 = not converted)
- Key Features:
 - Categorical: Lead Source, Last Activity, Lead Origin
 - Numerical: Total Time Spent on Website, Total Visits, Page Views

Data Cleaning and Preparation

- Dropped high-missing and low-informative columns (e.g., Lead Quality, Asymmetrique fields).
- Treated 'Select' as null; imputed missing values.
- One-hot encoded categorical variables.
- Scaled numerical features using StandardScaler.

Modeling Approach

- Model: Logistic Regression for interpretability and business alignment.
- Feature Selection: RFE (Recursive Feature Elimination) with top 20 variables.
- Training/Test Split: 70/30
- Evaluated using Accuracy, Precision, Recall, F1, ROC-AUC.

Model Performance

- Accuracy: ~80%
- Precision: ~85%
- Recall: ~73%
- F1 Score: ~78%
- ROC-AUC: ~87%
- Interpretation: High reliability in predicting lead conversion.

Top Predictive Variables

Numerical Features:

- Total Time Spent on Website
- Page Views Per Visit
- Total Visits

Top Categorical/Dummy Variables:

- Last Activity_Email Opened
- Lead Source_Organic Search
- Lead Origin_Landing Page Submission

Lead Scoring System

- Output: Probability Score (0 to 1), scaled to 0–100
- Leads segmented as:
 - Cold Leads: < 60
 - Warm Leads: 60–79
 - Hot Leads: 80+
- Used for prioritizing call strategy and nurturing plans.

Business Strategies for Different Phases

Aggressive Phase (Intern Support):

- Lower threshold to 60 to maximize reach
- Prioritize high-scoring leads for intern outreach

Efficiency Phase (Low Activity):

- Raise threshold to 90+
- Focus only on high intent leads
- Use automated emails for others

Conclusion & Recommendations

- Logistic model identifies key behaviours and boosts lead targeting.
- Recommend operationalizing lead scores into CRM.
- Periodically retrain model as behaviour patterns evolve.
- Next step: A/B testing lead prioritization strategy.